

LAB 5: GNSS data processing

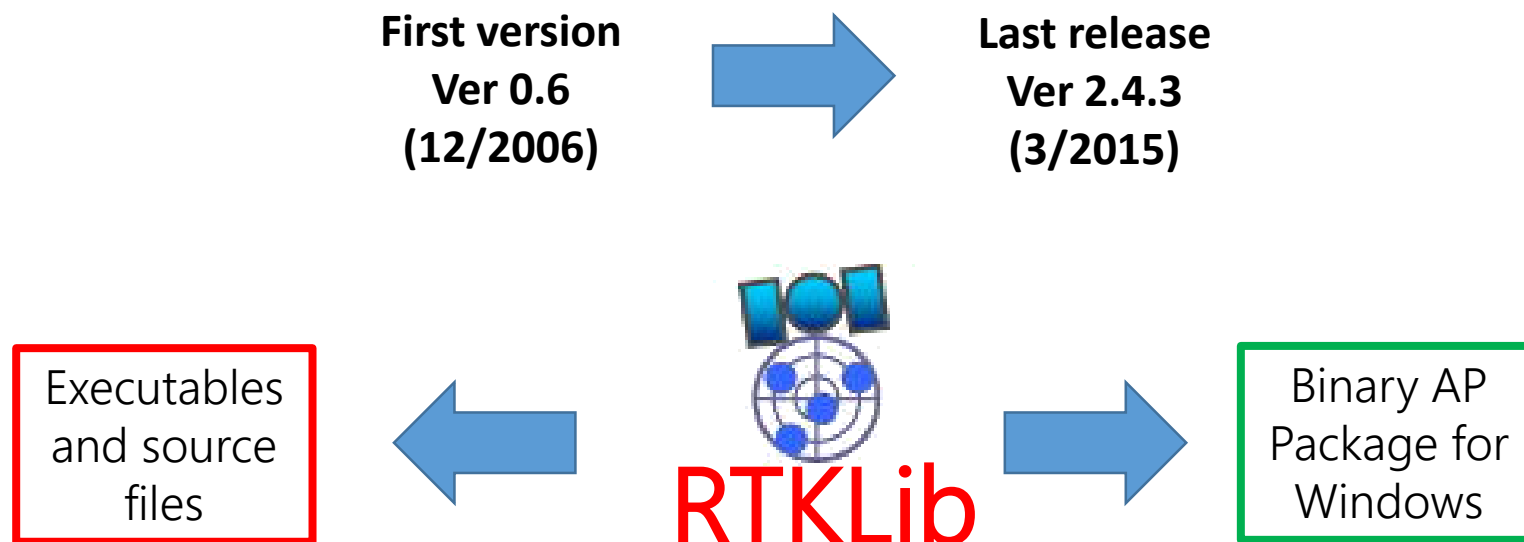




RTKLIB

<http://www.rtklib.com/>

RTKLIB is a suite composed of numerous tools and applications, developed by Tomoji Takasu of the Laboratory of Satellite Navigation, Tokyo University of Marine Science and Technology.



<http://wiki.openstreetmap.org/wiki/RTKLIB>





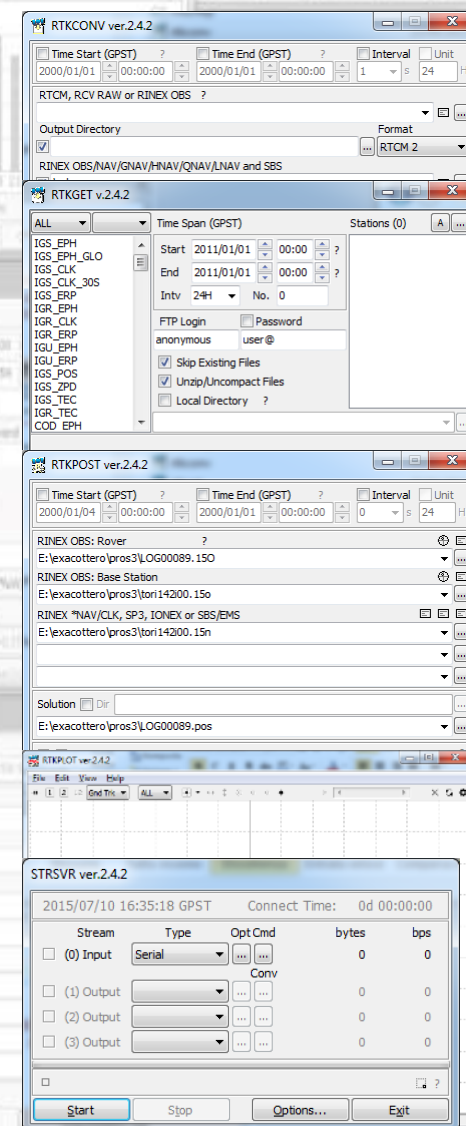
- **Supported satellite constellations:** GPS, GLONASS, GALILEO, BEIDOU, QZSS, SBAS
- **Types of positioning obtainable:** Single, DGPS / DGNSS, Kinematic, Static, Moving-Baseline, Fixed, PPP-Kinematic, PPP-Static and PPP-Fixed
- **Supported protocols:** RINEX 2.10, 2.11, 2.12 OBS / NAV / GNAV / HNAV / LNAV / QNAV, RINEX 3.00, 3.01, 3.02 OBS / NAV, RINEX 3.02 CLK, RTCM ver.2.3, RTCM ver.3.1 (with amendment 1-5), ver.3.2, BINEX, NTRIP 1.0, RTCA / DO-229C, NMEA 0183, SP3-c, ANTEX 1.4, IONEX 1.0, NGS PCV and EMS 2.0
- **External communications available:** Serial, TCP / IP, NTRIP, local log file (record and playback) and FTP / HTTP (automatic download)
- **Supported GNSS receivers:** NovAtel: OEM4 / V / 6, OEM3, OEMStar, Superstar II, Hemisphere: Eclipse, Crescent, u-blox: LEA-4T / 5T / 6T, SkyTraq: S1315F, JAVAD: GRIL / GREIS, Furuno: GW -10 II / III and NVS NV08C BINR



RTKLIB

RTKlib is made up of the following modules:

- RTKconv (data conversion)
- RTKget (download data and products from webserver)
- RTKpost (post-processing data)
- RTKnavi (RTK positioning)
- RTKplot (visualization of the GNSS solution)
- STRSVR (streaming data to server)





RTKLIB

RTKPOST is the tool dedicated to GNSS data processing.

RTKPOST ver.2.4.3 b33

☐ Time Start (GPST) ? ☐ Time End (GPST) ? ☐ Interval ☐ Unit

2000/01/01 00:00:00 2000/01/01 00:00:00 0 s 24 H

RINEX OBS ?

RINEX OBS: Base Station

RINEX NAV/CLK, SP3, FCB, IONEX, SBS/EMS or RTCM

Solution ☐ Dir

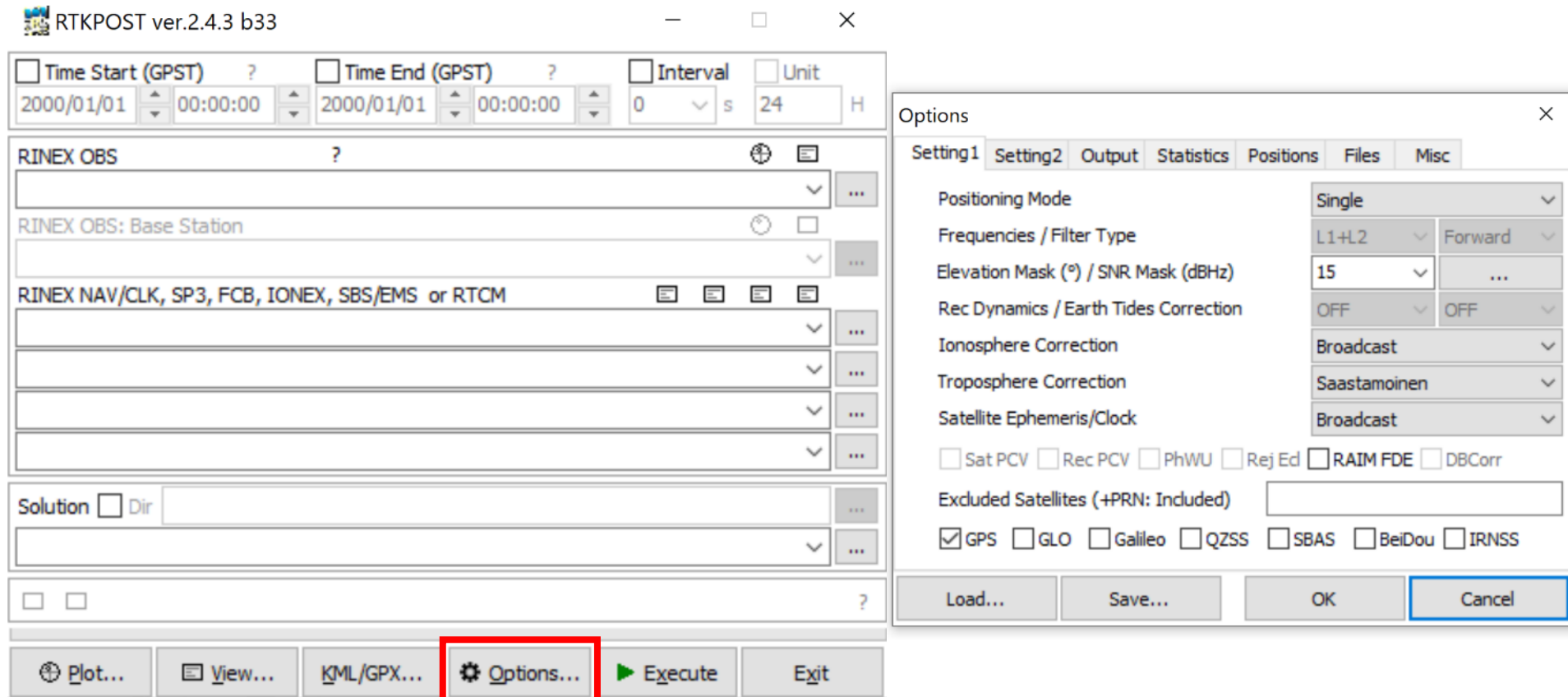
☐ ☐ ?

Plot... View... KML/GPX... Options... Execute Exit



RTKLIB

This tool allows to process GNSS data considering different positioning techniques and different frequencies





RTKLIB

It is possible to fix the phase ambiguity for GPS, GLONASS and Beidou satellites, in different ways:

- FIX and HOLD
- Instantaneous
- Continuous
- PPP-AR

It is also possible to consider only FLOAT solutions without seeking to fix the phase ambiguity.

Setting1	Setting2	Output	Stats	Positions	Files	Misc
Integer Ambiguity Res (GPS/GLO/BDS)	Fix ar	ON	ON			
Min Ratio to Fix Ambiguity	3					
Min Confidence / Max FCB to Fix Amb	0.9999	0.25				
Min Lock / Elevation (°) to Fix Amb	10	0				
Min Fix / Elevation (°) to Hold Amb	10	0				
Outage to Reset Amb/Slip Thres (m)	5	0.050				
Max Age of Diff (s) / Sync Solution	30.0	ON				
Reject Threshold of GDOP/Innov (m)	30.0	30.0				
Number of Filter Iteration	1					
☐ Baseline Length Constraint (m)	0.000	0.000				

Buttons: Load..., Save..., OK, Cancel





The fixing strategies for the GPS constellation are:

- **OFF:** No ambiguity fixing
- **Continuous:** ambiguities are continuously estimated and resolved in full
- **Instantaneous:** ambiguities are estimated and resolved era by era (epoch - by - epoch)
- **Fix and Hold:** ambiguities are continuously estimated and resolved in full. If the validation is OK (ratio test), the ambiguities are kept fixed at the values estimated at the previous time.
- **PPP - AR:** resolution of ambiguity in PPP (Experimental)



RTKLIB

The solutions can be exported in different formats

- Lat/Lon/Height
- X/Y/Z - ECEF
- E/N/U baseline
- NMEA 0183 message

It is possible to export both ellipsoidal heights and orthometric heights, considering different geoid models:

- EGM2008
- EGM96
- Internal model (low res)

The screenshot shows the 'Options' dialog box with the 'Output' tab selected. The dialog has several tabs: Setting1, Setting2, Output, Stats, Positions, Files, and Misc. The 'Output' tab contains the following settings:

Setting	Value
Solution Format	Lat/Lon/Height
Output Header/Processing Options	ON
Time Format / # of Decimals	hh:mm:ss GPST 3
Latitude / Longitude Format	ddd mm ss.sss
Field Separator	
Datum/Height	WGS84 Ellipsoidal
Geoid Model	Internal
Solution for Static Mode	Single
NMEA Interval (s) RMC/GGA, GSA/GSV	0 0
Output Solution Status / Debug Trace	OFF

At the bottom of the dialog are buttons for 'Load...', 'Save...', 'OK', and 'Cancel'.



RTKLIB

Particular attention must be paid to the insertion of the correct antenna model for both the master and the rover.

It is possible to constrain the coordinates of the master station in different ways:

- RINEX header position
- Lat/lon/height
- X/Y/Z-ECEF
- from file

Options

Setting1 Setting2 Output Stats Positions Files Misc

Rover

Lat/Lon/Height (deg/m) ...

-90.000000000 0.000000000 -6378137.0000

☐ Antenna Type (*: Auto) Delta-E/N/U (m)

0.0000 0.0000 0.0000

Base Station

RINEX Header Postion ...

-90.000000000 0.000000000 -6378137.0000

☐ Antenna Type (*: Auto) Delta-E/N/U (m)

0.0000 0.0000 0.0000

Station Position File

Load... Save... OK Cancel



RTKLIB

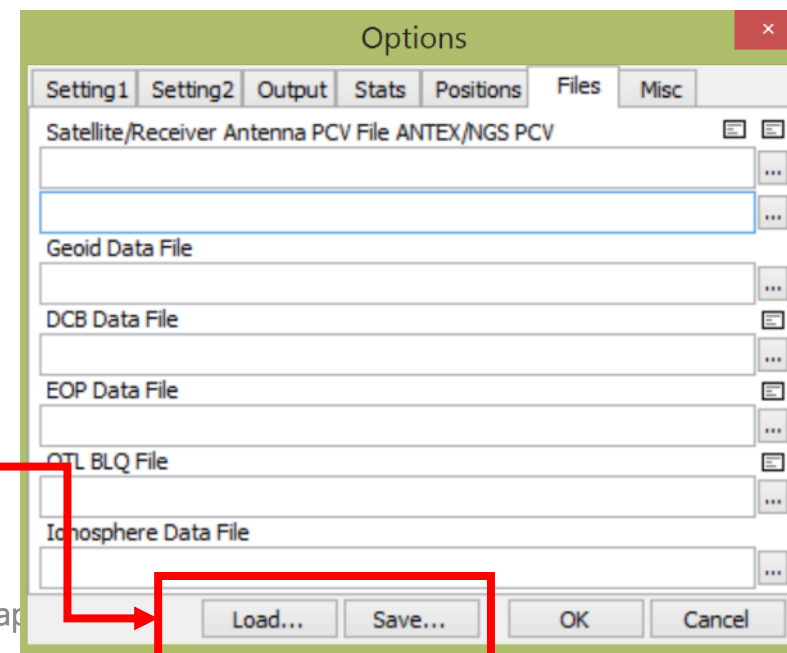
Other input files can be considered, especially for precision applications (e.g. geodetic points, CORSs ...):

- PCV for satellite antennas/rec
- DCB files
- EOP
- OTL BLQ
- Ionospheric delays

Which can be
downloaded with
RTKGET

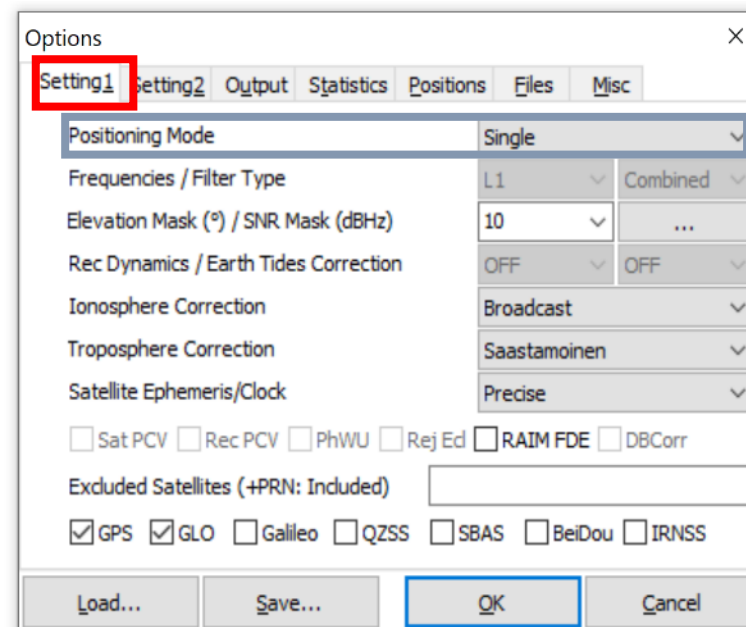
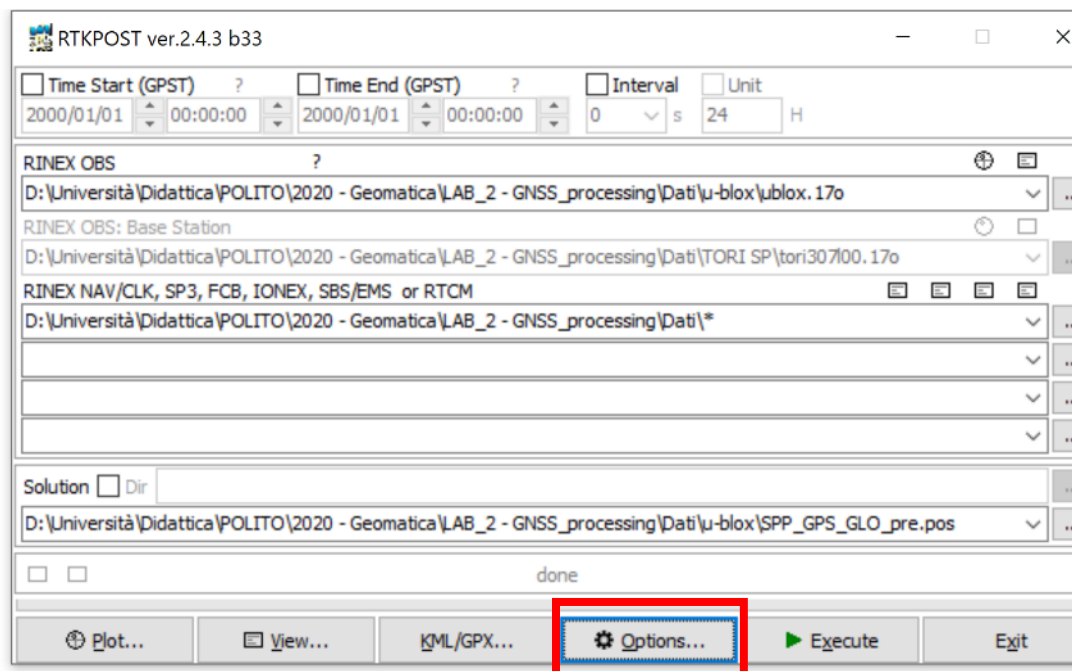
It is also possible to consider further
geoid models

Configuration files can be saved and
loaded to speed up processing

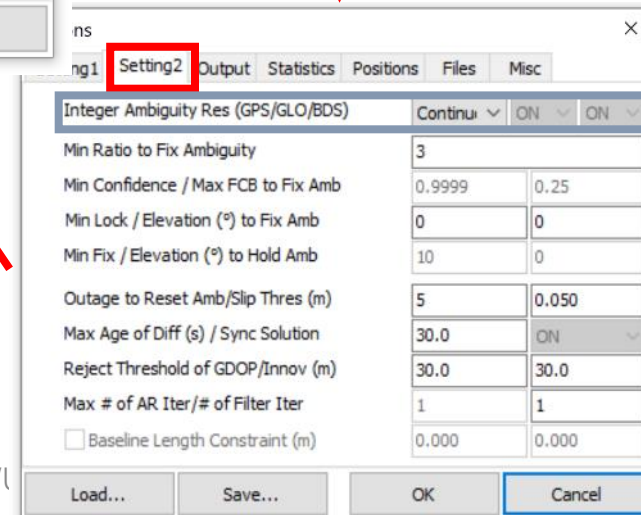
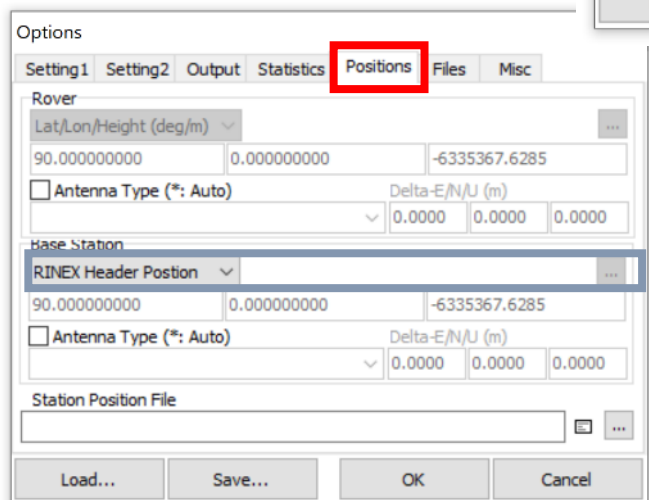
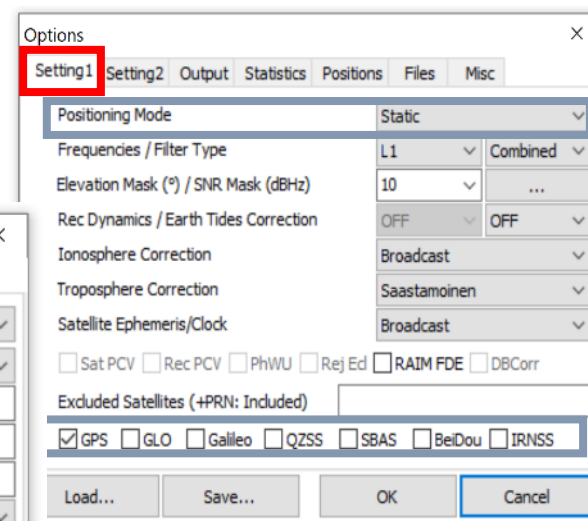
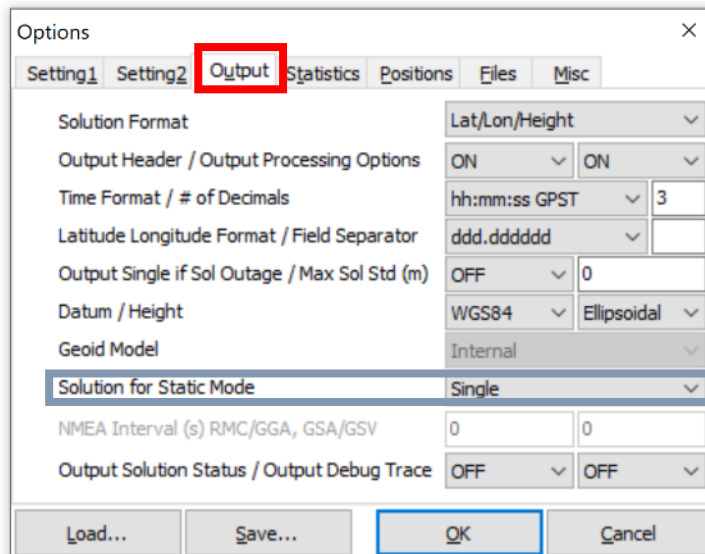
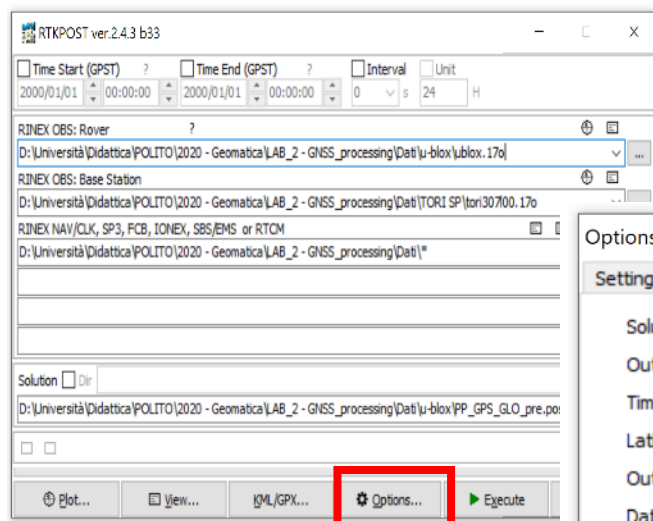




RTKLIB - SPP



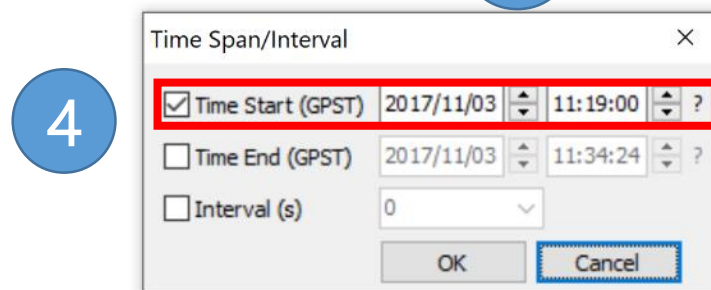
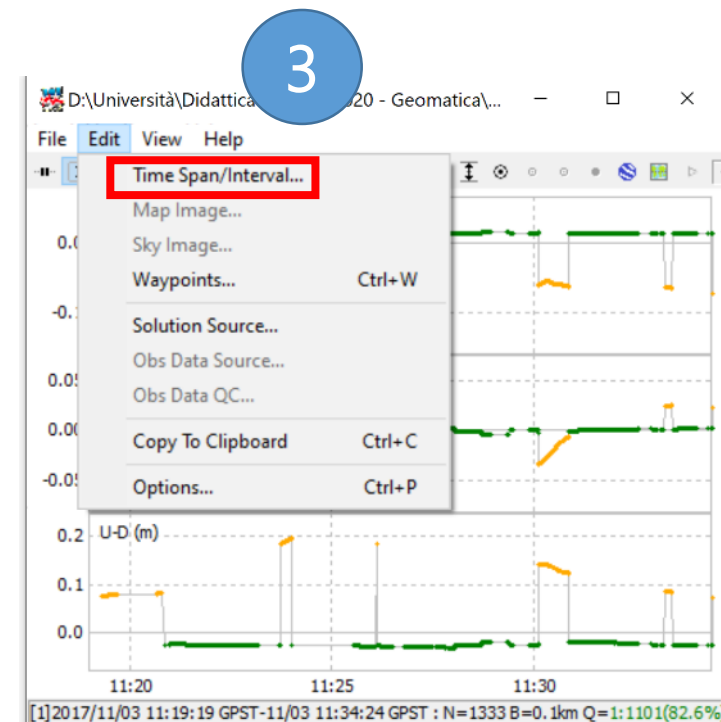
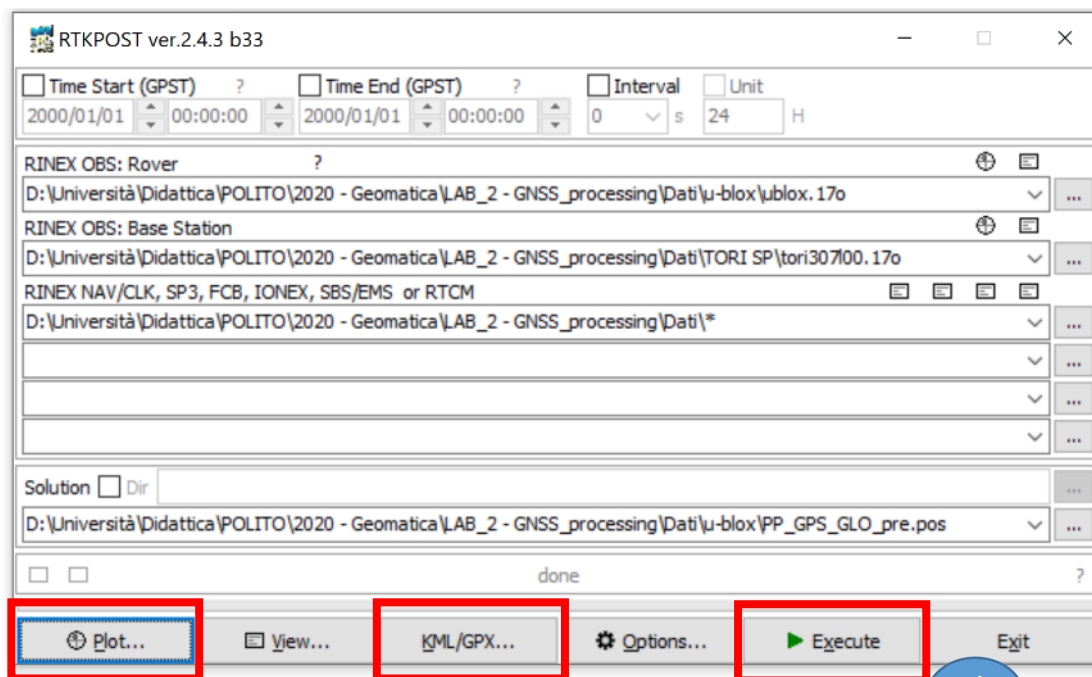
RTKLIB – relative positioning





RTKLIB – relative positioning

To view the results:





Ex 1. GNSS DATA PROCESSING (static)

INPUT: The available dataset is composed of:

- **Static data**

- A folder called **TORI** which contains:
 - tori119k45.25o → observation file (code and phase measurements)
 - tori119k45.25n → GPS navigation file
 - tori119k45.25g → GLONASS navigation file

INPUT DATASET: GNSS
STATIC.zip

Reference Station
TORINO
(ETRF2000)

UTM Nord (m):	4990861.136	Lat:	45° 03' 48.112513" N
UTM Est (m):	394604.567	Long:	7° 39' 40.599491" E
Qta ort. (m):	261.67	h ell. (m):	310.74

- Data collected with MASTER station which contains:
 - GPS01190.25o
 - GPS01190.25n

GNSS ROVER
 $hS = 1.117 + 0.36 = 1.477m$

- 1) Estimate the solution as «single solution»
- 2) Estimate the solution as «all epochs» and comment the results (quality, ratio, num sat...)
- 3) Compare the solution with the official reference





Ex 2. GNSS DATA PROCESSING (kinematic)

INPUT: The available dataset is composed of:

- **Kinematic data**

- A folder called **MONV SP** which contains:
 - monv020i00.20o
 - monv020i00.20n
 - monv020i00.20g
- A folder called **SnowGroomer** which contains:
 - 20200120_COM3_165803.20o
 - 20200120_COM3_165803.20n
 - 20200120_COM3_165803.20g

INPUT DATASET: GNSS
KINEMATIC.zip

Reference Station
MONDOVî

Kinematic data

Kinematic: 2020/01/20 17:00:00





GOALS

GOALS:

1_A) Single point positioning of MASTER STATION

1_B) Relative Static positioning

- **MASTER post processing using TORI as Ref (GPS only)**
- **MASTER pp using TORI as RF (GPS+GLONASS)**
- **Results comparison, considering the single solution or “all epochs”**

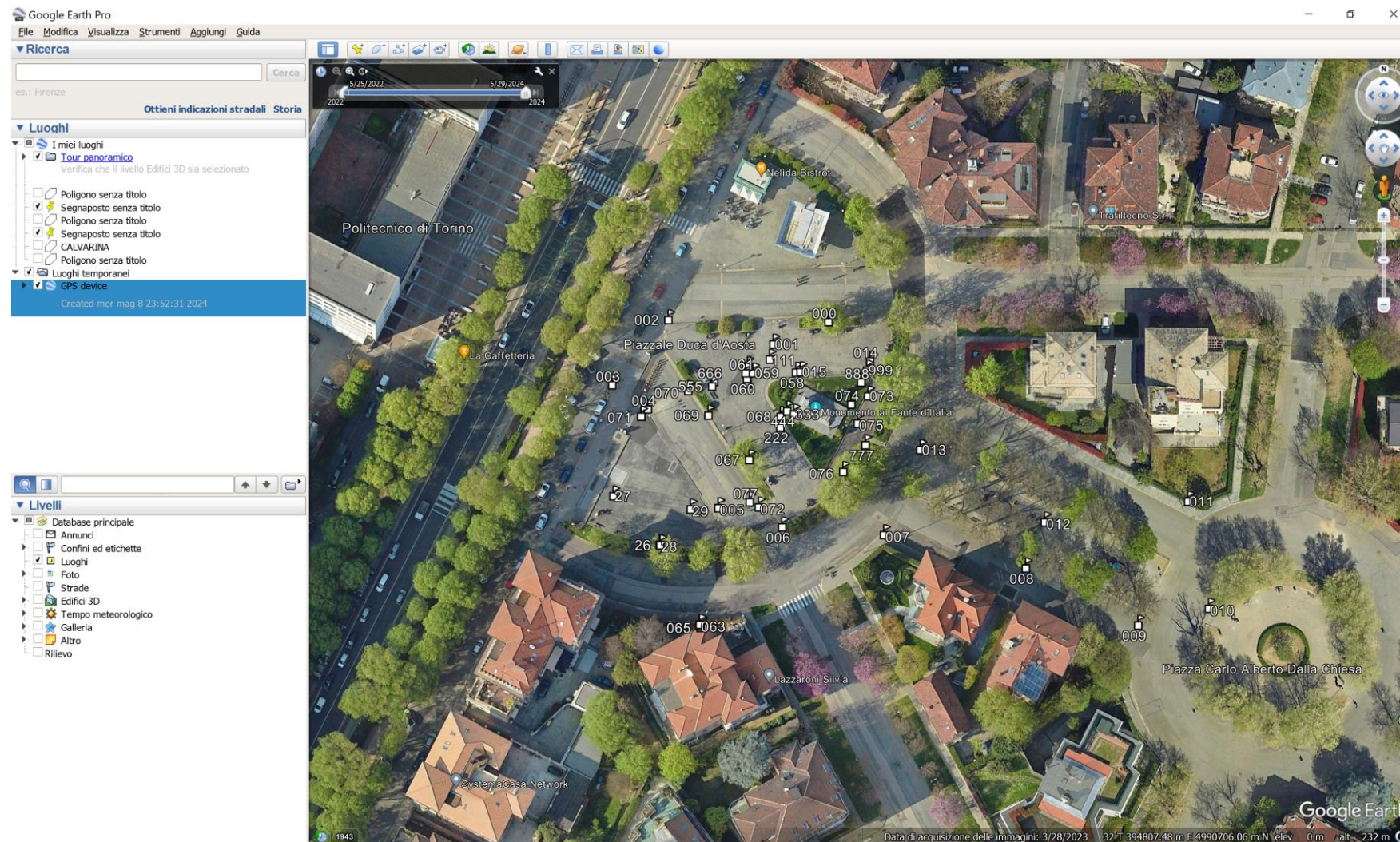
2_C) Kinematic positioning

- **Snow groom data pp using MONV as Ref (FIX & Hold)**
- **Snow groom data pp using MONV as Ref (Continuous)**
- **Results comparison**

3_D) Satellite analysis (Visibility, DOP, etc)



Ex 3. Survey in the field



GOAL:

- 1) To compare the coordinates collected with GNSS RTK solution and GARMIN solution and smartphone solution (if available)
- 2) To compare distances, considering the “reference” measured by tape.